The remembrance stop in traffic data: Istanbul at 09:05 on 10 November, 2020–2024

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Abstract

Every 10 November at 09:05 sirens sound across Turkey for Atatürk, and many drivers stop their vehicles on the road. We ask how much this slows Istanbul's traffic. We use the municipality's open data on hourly speeds in about 2 305 one-kilometre cells for 2020–2024. For each cell we compare the 09:00 hour with the hours before and after it, and we compare 10 November with 74 other dates of the same weekday in the surrounding sixteen weeks. Significance is judged against all 983 040 ways of picking one date per year as the event. Across all cells the 09:00 hour was slower on 10 November by the equivalent of 0.18 minutes of full stop per vehicle ($p = 0.116$), which is not distinguishable from an ordinary date. In cells that normally flow at 50–70 km/h the 09:00 hour lay below the line between its neighbours in all five years, by the equivalent of 0.64 minutes (90% interval 0.26 to 1.05; $p = 0.003$, and 0.015 after adjustment for testing four speed classes). Part of this gap arises because the 10:00 hour was unusually fast on 10 November. Against hours that do not adjoin the siren, the 09:00 hour was slower by 0.43 minutes ($p = 0.101$). The lowest recorded speed in the hour fell by 4.7%, while the highest speed and the number of vehicles did not change. On slow streets, cells in neighbourhoods with a higher vote for the CHP slowed more at 09:00 in all five years ($p = 0.0007$). The data contain no stationary vehicles, so these figures bound the true delay from below.

Keywords: collective ritual; commemoration; traffic speed; probe data; placebo test; Istanbul

1 Introduction

At 09:05 on 10 November, the minute of Mustafa Kemal Atatürk's death in 1938, sirens sound across Turkey. People stand in silence, and on the roads many drivers stop where they are. The custom is as old as the commemoration. On the first anniversary, in 1939, trams and vehicles in Istanbul stood still for five minutes at 09:05 (Uzun, 2019, p. 142). Official mourning ended in 1988, but the silence, the horns and the sirens continued, according to the historian of the ceremonies, without any obligation (Uzun, 2019). In a 1988 newspaper poll, 89.7% of respondents said that the standing silence at 09:05 should be kept (Uzun, 2019, p. 159).

How many people take part, and for how long, has never been measured. The scholarship on the commemoration is historical and ethnographic (Uzun, 2019; Özyürek, 2004; Goldman, 2018). The same holds for comparable rituals elsewhere: Vinitzky-Seroussi and Teeger (2010) describe drivers on Israeli highways stopping and standing by their cars during the memorial sirens, but report no count. Where economists have studied national days, they have lacked data on participation; Madestam and Yanagizawa-Drott (2011) proxy attendance at Fourth

of July celebrations with rainfall. Shared national moments can matter: Depetris-Chauvin et al. (2020) find that victories of national football teams strengthen national over ethnic identification in Africa. Surveys record what people say should be done. A synchronized stop on the road, however, leaves a trace in traffic data, and that trace is behaviour.

We estimate the size of this trace in Istanbul. The data are the hourly speed records that the Istanbul Metropolitan Municipality (İBB) publishes for cells of roughly one kilometre, from January 2020 to January 2025, covering five commemorations. The design has two steps. Within each cell and day we compare the 09:00 hour with the mean of the 08:00 and 10:00 hours, which removes anything that shifts a whole morning. We then compare this contrast on 10 November with the same contrast on other dates of the same weekday within eight weeks before and after, leaving out public holidays and the school break. To judge significance we ask where 10 November falls among all ways of choosing one date per year as "the event". The cells are strongly correlated within a day, so the effective sample is the number of dates, not the number of cells; the comparison across dates respects this. We express the slowdown as *full-stop-equivalent minutes*: the number of minutes for which every vehicle would have to stand still to lower the hour's mean speed by the estimated amount. If every vehicle stood still for the one to two minutes of the silence, and the data recorded it, the 09:00 hour would be slower by 1.7% to 3.4%. We expected less, because participation is partial and because, as we show, the data discard stationary vehicles.

We find a disturbance that is clear in one part of the network and absent from the rest. In cells that normally flow at 50–70 km/h at 09:00, the hour lay below the line between its neighbours on 10 November in all five years, by the equivalent of 0.64 minutes of full stop ($p = 0.003$; 0.015 after adjusting for the four speed classes examined). Averaged over all cells the estimate is 0.18 minutes, within the range of ordinary dates ($p = 0.116$). Congested cells and motorway cells show nothing. We find no difference between the European and Asian sides. District vote shares show no gradient, but neighbourhood vote shares do: on slow streets, where drivers are more likely to be local, cells in neighbourhoods with a higher CHP vote slow more at 09:00, in all five years.

Two points frame these numbers. First, the contrast responds both to a slow 09:00 hour and to a fast neighbouring hour, and on 10 November the 10:00 hour was itself fast in these cells. Measured against hours that do not adjoin the siren, the 09:00 hour was slower by 0.43 minutes, in 3 of 5 years, and this difference alone is not statistically significant ($p = 0.101$). The lowest speed recorded in the 09:00 hour fell by 4.7% while the highest speed and the number of vehicles were unchanged, which is what braking and restarting would produce, but this too is clearer against the later hour than against the earlier one. Second, the data contain no stationary vehicles: the lowest speed on record is 1 km/h from 2021 and 6 km/h in 2020. A vehicle that stands still drops out of the average, and what remains is braking, crawling and restarting around the stop. The estimates are therefore lower bounds on the delay.

The paper makes three contributions. It gives the first estimate, to our knowledge, of how a national moment of silence appears in traffic data, with a counterfactual and formal inference, and it states how far that estimate can be trusted. It documents properties of a widely used open data set that matter for anyone who uses it to study brief events, including an undocumented change in its speed threshold. And it shows what a wide pool of placebo dates changes relative to the common "same weekday, adjacent weeks" design: one finding of our own pilot disappears, and the control date one week after the event turns out to lie in a school holiday.

2 Background and related literature

2.1 The 09:05 silence

The 1939 programme of the ruling party set out a five-minute silence for party branches, schools, the press and radio; it said nothing about streets, sirens or vehicles, and yet the newspapers of the next day reported traffic standing still (Uzun, 2019). Car horns and the sirens of factories, ferries and ships are reported from the 1950s. The practice survived the end of official mourning in 1988, and Uzun (2019) describes daily life as almost coming to a halt at that minute, by people's own will. He cites no rule that obliges drivers to stop.

Two accounts of Atatürk commemoration since the 1990s bear on who takes part. Özyürek (2004) describes the move of Atatürk imagery from state offices into homes and shops, taken up by secular, middle-class, long-settled urban citizens who stressed that their devotion was freely chosen; she also warns that presenting support as voluntary can mask the power of the state. Goldman (2018) argues that commemoration became more inclusive after 2013 and that, after 2016, secularist and conservative participation became hard to tell apart. The first account suggests more participation in secular districts; the second suggests little difference. We return to this in Section 5.6. In the terms of Bellah (1967), an annual rite of this kind is part of a civil religion, a set of beliefs, symbols and rituals about sacred things that is institutionalized in a collectivity; whether Kemalism is better described as a civil or a political religion is debated (Karakuş, 2023). Vinitzky-Seroussi and Teeger (2010) classify the moment of silence as overt silence in the service of memory, most effective when a nationwide siren marks it and a norm to stand still accompanies it, and note that compliance is internalized, not policed.

2.2 Collective acts in passive data

A small literature infers the size or timing of collective acts from data collected for other purposes. Its standard counterfactual is the same time slot in other weeks: telephone calls after emergencies (Bagrow et al., 2011), mobile-network activity during football matches (Pintér and Felde, 2021), and traffic counts around a solar eclipse (Ngeni et al., 2022). Inference in these studies is informal, based on thresholds or bands around a handful of baseline days. The closest analogue to our question is Earth Hour, a scheduled and voluntary act: Olexsak and Meier (2014) compile 274 reported changes in electricity demand, averaging a 4% reduction, and note that the answer depends on the baseline and that a single earlier week is not an adequate one; they also give an example of demand rebounding after a synchronized dip. Converting a signal into a head count requires ground truth, as Botta et al. (2015) show with stadium attendance. We report our estimate in minutes per vehicle and leave it in that unit.

Public acts have also been counted directly. Dinas et al. (2024) count national flags street by street in Madrid, and Parravano et al. (2015) count independence flags on Barcelona facades. Both find that display responds to what others nearby do, and Dinas et al. (2024) find that public behaviour changed while private preferences did not. Skitka (2005) finds that self-reported flag display after September 2001 tracked patriotism more than hostility to out-groups. These studies caution against reading a public act as a private preference. The caution applies with more force on the road, where a stopped car physically stops the cars behind it.

2.3 One-day events and the choice of control days

The design closest to ours compares an event day with the same weekday one week earlier and one week later, in the same clock hours. It was used for road deaths on Super Bowl Sunday (Redelmeier and Stewart, 2003), on United States presidential election days (Redelmeier and Tibshirani, 2008), on tax day (Redelmeier and Yarnell, 2012), on 20 April (Staples and Redelmeier, 2018) and during the 2017 eclipse (Redelmeier and Staples, 2024). Two reanalyses show its weak point. Zhang and Aronow (2016) compared election days with 200 other Tuesdays instead of two: the estimate was unchanged, but its p-value rose from below 0.001 to 0.07, and the original test rejected on 43% of ordinary Tuesdays. Harper and Palayew (2019) widened the control set for 20 April and found that the estimate moved towards zero and its interval included no effect. Smith (2016) uses placebo dates in a permutation test for the effect of daylight saving time on fatal crashes, excluding dates that the event itself contaminates. We follow these papers: our reference distribution comes from placebo dates, the pool is wide, and its rules are fixed before estimation. A related strand identifies effects from their timing within the day: family violence rises in the hours after an unexpected football loss (Card and Dahl, 2011), and reports of assault rise on college game days but not on other autumn Saturdays (Lindo et al., 2018). We use the same logic when we move the contrast to other hours and other days.

Seasonal drift is a separate danger. Anderson (2014) reports that a before-and-after comparison on unaffected Los Angeles freeways produced a spurious 32% rise in delay, because delays trend upward through October and November. Our event always falls in the same week of the same month, so we use placebo dates on both sides and check trend-adjusted versions.

2.4 Traffic shocks and probe measurement

Economists have measured the traffic effects of shocks that last from a day to several weeks: transit strikes in Los Angeles (Anderson, 2014), Germany (Bauernschuster et al., 2017) and Rome (Adler et al., 2021), and the lifting of a carpool rule in Jakarta (Hanna et al., 2017). These studies compare the same clock hour across days and report effects from about 2% to more than 80% of travel time. Speeds from phone-based probes also register holidays, processions and strikes in Indian cities, and Sundays are about 4.5% faster than working days (Akbar et al., 2023). None studies a shock shorter than an hour, and none compares an hour with its neighbours on the same day. Gulek (2024) is the closest study for Turkey: it uses the timing of the Ramadan fast to identify the effect of hunger on traffic accidents, and uses İBB's traffic index as a check on exposure.

What a speed average measures depends on how it is formed. Wardrop (1952) distinguished the time-mean speed, an average over vehicles passing a point, from the space-mean speed, an average over vehicles on a stretch of road, and showed that the first exceeds the second by the variance of speeds divided by their mean. He also distinguished running speed, which excludes stops, from journey speed, which includes them. The gap between time-mean and space-mean speed is a few per cent in free flow and 10–30% in congestion (Rakha and Zhang, 2005; Knoop et al., 2009). Probe data add a further choice between sampling vehicles at fixed time intervals and sampling them when they cross a line; time sampling returns more low-speed observations (Herrera et al., 2010). Speed measures built from GPS traces often drop trips that contain stops (Kreindler, 2024). The theory of kinematic waves describes what follows a stoppage: the queue discharges at capacity for a time that grows steeply as the arriving flow approaches

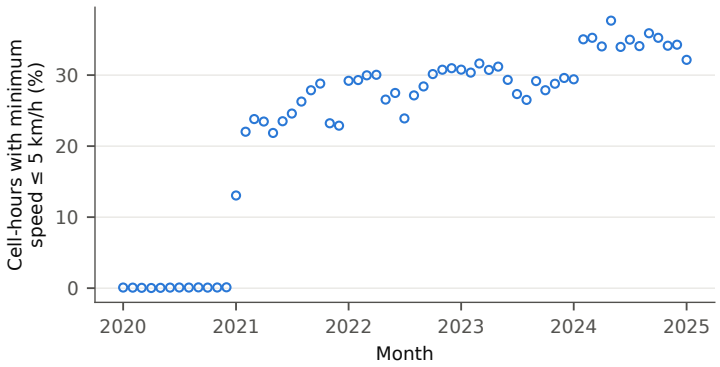


Figure 1: Share of cell-hours whose minimum recorded speed is 5 km/h or less, by month. The threshold below which observations are discarded fell from about 6 km/h to 1 km/h in January 2021.

capacity, and each start costs several seconds (Lighthill and Whitham, 1955; Richards, 1956). How steeply speed falls with density has been estimated causally for Beijing by Yang et al. (2020). Adler et al. (2021) note that hourly data conceal variation within the hour. Section 4.2 sets out what each of these choices implies for a brief stop.

3 Data

3.1 Hourly speeds by cell

The İBB open data portal publishes the Hourly Traffic Density Data Set as one file per month from January 2020 to January 2025.[1] Each row gives, for one geohash cell of six characters (about 1.2 by 0.6 km) and one hour, the average, minimum and maximum speed in km/h and a number of vehicles. We downloaded 61 monthly files, about 100 million cell-hours. Around 2 305 cells report in a typical hour. Published users of the data describe the same columns (Utku, 2023; Özkurt et al., 2025; Tanberk et al., 2023); related İBB speed products appear in Olug et al. (2024) and Alemdar et al. (2021). These sources and the portal describe the columns. The rule by which speeds within the hour are averaged is not published, and Section 4.2 derives what each candidate rule implies for a stop. The portal defines that number as the count of distinct vehicles in the cell in the hour; depending on the year, the median cell records between 38 and 67 of them at 09:00, so they are a sample of traffic, not a census.

The data reveal one undocumented rule. The lowest recorded speed is never zero. Until December 2020 almost no cell-hour has a minimum speed below 6 km/h (0.1% of cell-hours at or below 5 km/h); from January 2021 about 29% do, and the lowest value is 1 km/h (Figure 1). Observations of stationary or nearly stationary vehicles are therefore removed before averaging, with a threshold that changed at the start of 2021. Sample rows reproduced in Tanberk et al. (2023) are consistent with the two floors. The consequence for a stop is direct: a vehicle standing still contributes nothing to the average.

[1] İBB Açık Veri Portalı, `https://data.ibb.gov.tr/dataset/hourly-traffic-density-data-set`. The data are published under the İBB Open Data License. Atıf 4.0 Uluslararası (CC BY 4.0) kapsamında lisanslanan kamu sektörü bilgilerini içerir.

Table 1: Sample by year. Cells are those observed on 10 November and on at least three quarters of the pool dates. The lowest recorded speed is the first percentile of the minimum speed at 09:00.

Year	Weekday of 10 November	Dates in the pool	Cells	Median probe vehicles, 09h	Lowest recorded speed (km/h)
2020	Tuesday	16	2 277	66	6
2021	Wednesday	16	2 331	64	1
2022	Thursday	16	2 310	67	1
2023	Friday	16	2 285	38	1
2024	Sunday	15	2 324	40	1

3.2 Event dates and placebo dates

The five commemorations fell on a Tuesday (2020), Wednesday (2021), Thursday (2022), Friday (2023) and Sunday (2024). For each year the *pool* contains 10 November and every other date of the same weekday within eight weeks before and after. We remove public holidays, including the half day on 28 October, and the week of the schools' autumn break, which in these years began on the Monday after 10 November.[2] The break matters because the date one week after the event, the natural control in an adjacent-weeks design, falls inside it in four of the five years. These rules were fixed before estimation. Table 1 gives the resulting sample: 79 dates, of which 74 are placebo dates.

3.3 Other data

Hourly precipitation and temperature for central Istanbul come from the Open-Meteo archive of reanalysis weather. District boundaries come from OpenStreetMap, and district results of the 2024 metropolitan mayoral election from the tallies compiled in the Wikipedia article "2024 Istanbul mayoral election". Neighbourhood (mahalle) results come from TÜİK's election statistics for two elections: the 14 May 2023 parliamentary election, with party-list votes for 962 neighbourhoods, and the 31 March 2024 metropolitan mayoral election, with votes for 960. We use the parliamentary vote as the main measure, because it records party choice; the 2024 mayoral vote also reflects the candidates and serves as a check. The CHP shares of the two elections correlate at 0.94 across neighbourhoods. Neighbourhood boundaries come from the land-registry (tapu kadastro) polygons; they follow the administrative neighbourhoods closely but not exactly. Each cell receives the vote share of the neighbourhoods it overlaps, weighted by area. We keep the 3 030 cells whose area is at least half covered by neighbourhoods with a matched vote record (median coverage 100%); across these cells the 2023 CHP share has a standard deviation of 12.1 points. Finally, İBB's traffic index web service returns a city-wide congestion index (1–99, higher is more congested) at five-minute intervals for the latest 60 days. We saved the response on the evening of 10 November 2026; it covers 11 September to 10 November 2026, and Section 5.7 uses it to show what a city-wide average hides and the cell-level design recovers. The hourly version of this index has been used by Alcan and Kaçar (2022) and Gulek (2024), who describes it as built from camera-measured speeds on main roads.

[2] Ministry of National Education calendars: 16–20 November 2020, 15–19 November 2021, 14–18 November 2022, 13–17 November 2023 and 11–15 November 2024. The data agree: on the break-week date, speeds at 07:00–09:00 were 6–7% higher than on the neighbouring same-weekday dates in 2021–2023.

4 Method

4.1 Contrast, statistic and estimate

Let $v_{gyd}(h)$ be the average speed in cell g on date d of year y in the hour starting at h. The within-day contrast is

$$c_{gyd} = \ln v_{gyd}(9) - \tfrac{1}{2}\left[\ln v_{gyd}(8) + \ln v_{gyd}(10)\right]. \tag{1}$$

It is zero when the 09:00 hour lies on the line between its neighbours and negative when it dips below. Anything that scales all three hours alike, such as rain all morning or a quiet day, cancels. For each year we keep the cells G_y observed on 10 November and on at least three quarters of the pool dates, subtract each cell's mean contrast over the pool, and average over cells:

$$M_{yd} = \frac{1}{|G_y|} \sum_{g \in G_y} \left(c_{gyd} - \bar{c}_{gy}\right). \tag{2}$$

All further steps use only these date-level numbers. The statistic for date d is its difference from the other $K_y - 1$ dates of its year, and the estimate is the mean over years of the statistic on the event date e_y:

$$T_y(d) = M_{yd} - \frac{1}{K_y - 1} \sum_{d' \neq d} M_{yd'}, \qquad D = \frac{1}{5} \sum_{y} T_y(e_y). \tag{3}$$

Years are weighted equally because each contributes one event.

Cells are grouped by their usual speed at 09:00, the mean over all pool dates of the year, into four classes: below 30, 30–50, 50–70 and 70 km/h or more. The classes were set in a pilot analysis that used three control dates per year. The grouping uses every pool date, the event date included, so that it does not depend on which date is called the event. Section 6 reports a version that classifies cells from other days altogether.

4.2 Full-stop-equivalent minutes

We report $m = 60\,(1 - e^{D})$. If each vehicle's speed is sampled at regular intervals and all samples are averaged, the hourly mean is distance travelled divided by time, and a stop of k minutes by a share π of vehicles lowers it by the factor $1 - \pi k/60$. Then $m = \pi k$: the minutes of standstill per vehicle. The assumption fails here, because samples below the threshold are discarded. Suppose a vehicle brakes for t_d minutes, stands for s and accelerates for t_a. With stationary samples removed, the average becomes distance over *moving* time, which is Wardrop's running speed, and

$$m \approx \pi\, \frac{t_d + t_a}{2}\,: \tag{4}$$

the standing time s disappears, and only the braking and restarting register. If speeds are recorded when vehicles cross a point instead of at fixed intervals, the loss is smaller still, about $\pi(t_d + t_a)/6$, because by Wardrop's identity a flow-weighted mean is blind to a full stop. With ordinary braking and acceleration from 60 km/h, $t_d + t_a$ is about half a minute. These cases are our own derivation from the definitions in Wardrop (1952) and Herrera et al. (2010); the appendix gives the steps. Vehicles that slow to a crawl without stopping stay in the average under every scheme.

We therefore read m as a lower bound on the minutes of delay per vehicle, and we use three features of the data to check the reading: the minimum speed in the hour should fall; the maximum speed should not; and the number of distinct vehicles should not change.

4.3 Inference

The estimate rests on five event dates. However many cells are observed, the uncertainty in such a design comes from shocks at the level of the date and does not shrink as cells are added (Conley and Taber, 2011; Ferman and Pinto, 2019). Standard errors computed from the cells, or clustered on the date when only a handful of dates are treated, reject far too often, and bootstrap corrections fail when a single cluster is treated (MacKinnon and Webb, 2016, 2018; Bertrand et al., 2004). That all cells of the city are observed does not remove the uncertainty, which comes from which dates carry the event (Abadie et al., 2020). Reanalyses of published experiments show how much conventional inference can overstate significance in designs with few effective observations (Young, 2018). In our data the variance of a date's mean contrast is 4.7 times what independent cells would give.

We therefore compare the event dates with placebo dates. Consider every assignment that picks one date from each year's pool. For each we compute the mean over years of T_y, and the p-value is the share of assignments whose value is as low as D or lower, the observed one included. There are 983 040 assignments and we enumerate them all. This is the logic of a randomization test with years as blocks (Rosenbaum, 2002b; Ernst, 2004; Heß, 2017; Athey and Imbens, 2017), of placebo tests in comparative case studies (Abadie et al., 2010, 2014; Firpo and Possebom, 2018), and of placebo tests over clusters, which may be stretches of time (Hagemann, 2019). The quantity estimated is the average effect realized on the five event dates (Alvarez and Ferman, 2023). It is exact if, absent the siren, each date of a pool were equally likely to produce any of the observed values. The date of the commemoration is fixed by the calendar, not drawn at random, so this is an assumption, and the p-value is best read as the significance of 10 November against a uniform benchmark over comparable dates (Abadie, 2021; MacKinnon and Webb, 2020). It allows any dependence between cells within a date. It rules out a trend across the weeks of the pool and dependence between neighbouring weeks. Permutation over time is exact when the data are exchangeable under the null, and it over-rejects when a trend is left in the comparison (Chernozhukov et al., 2021; Roth et al., 2023); we therefore check trend-adjusted versions. Because the test is one-sided and uses a difference in means, it remains valid for the wider null hypothesis that the siren raised speed on no date (Caughey et al., 2023).

For intervals we invert the test: we subtract a hypothesized constant effect from the event dates and keep the values that are not rejected at the 5% level in either direction, which gives a 90% interval (Rosenbaum, 2002b,a). Without a constant effect the interval's lower end bounds the largest yearly effect (Caughey et al., 2023).

Regression version. The comparison above uses only same-weekday dates. A fixed-effects regression uses every day within eight weeks of the event and adds controls. For each year we regress $\ln v_{gdh}$, hours 06 to 12, on cell-by-date effects, which absorb anything that moves a cell's whole morning; cell-by-hour effects for each day type (Monday to Thursday, Friday, Saturday, Sunday), which give each cell its usual hourly profile; hour effects for school-break days; and hourly rainfall, a wet-hour indicator and temperature. The fixed effects and controls

are partialled out without the event, and the event statistic is formed from the residuals of the same-weekday pool dates exactly as above. Under the null of no effect this is the covariance-adjusted randomization test of Rosenbaum (2002a), so the placebo-date inference carries over unchanged.

The residuals are heavy-tailed, and most of the extreme values come from cells observed through few probe vehicles. Down-weighting extreme residuals one by one, as robust M-estimation does, would also down-weight cell-hours in which vehicles stopped, because a stop is itself an extreme residual. We therefore weight cells, not observations. Each cell receives a fixed weight equal to the inverse of its residual variance, estimated only on days outside the same-weekday pool, so that neither 10 November nor any placebo date informs its own weight. This is inverse-variance weighting of the kind Ferman and Pinto (2019) use to correct heteroskedasticity across groups in designs with few treated units. A cell that is quiet on ordinary days keeps its full weight on 10 November, however large its response. Date-level statistics are precision-weighted means of the cell residuals, and this is the main regression version. The regression also gives each cell's response, which we regress on the local vote share for each date, by weighted least squares with the same weights, to test for a political gradient over the same assignments.

Four speed classes are tested, so we report p-values adjusted by the stepdown procedure of Romano and Wolf (2005b), computed from the joint distribution of the four class statistics over the same assignments. It controls the probability of any false rejection under the same assumption as the single tests, and is less conservative than the Bonferroni and Holm corrections when the statistics are positively dependent (Romano and Wolf, 2005a). A test of "no effect in a class" does not test whether classes differ (List et al., 2019; Ding et al., 2015), so we also test differences between classes directly, over the same assignments. As a check that does not rely on comparing dates, we apply the sign-change test of Canay et al. (2017) to the five yearly estimates; with five years its smallest attainable one-sided p-value is $1/32 = 0.031$. All computations use Python 3.12 with NumPy 2.3 and pandas 2.3. Assignments are enumerated exactly when there are fewer than six million of them; for the largest pool, two million are drawn at random.

5 Results

5.1 Main estimates

Table 2 gives the estimates. Over all cells the effect on speed in the 09:00 hour is -0.30 log points ($\times 100$), or 0.18 full-stop-equivalent minutes, with a 90% interval from -0.07 to 0.43 minutes. The estimate is negative in 3 of 5 years and its p-value is 0.116. A slowdown of this size is what ordinary dates produce.

The picture differs by class. In cells that usually flow at 50–70 km/h the effect is -1.07, or 0.64 minutes (90% interval 0.26 to 1.05), and it is negative in all five years (Figure 2). Only a fraction 0.003 of assignments give a value this low; the stepdown-adjusted p-value is 0.015, and a joint test of the four classes gives 0.015. The sign-change test on the five yearly estimates gives 0.031, its minimum. The other three classes are indistinguishable from ordinary dates: 0.04 minutes below 30 km/h, 0.26 at 30–50 km/h and 0.01 at 70 km/h or more. The differences between the 50–70 class and these three have one-sided p-values of 0.039, 0.093 and 0.006. The yearly estimates for this class range from 0.18 minutes in 2020 to 1.12 in 2024 (Appendix

Table 2: Effect of the 10 November siren on speed in the 09:00 hour, 2020–2024. "Effect" is D, the mean over years of the event date's contrast minus that of the placebo dates, in log points ×100. Full-stop minutes are $60(1 - e^{D})$. The interval inverts the two-sided placebo test. p-values are one-sided shares of 983 040 assignments of one date per year; the stepdown adjustment covers the four speed classes.

Usual 09h speed	Cells	Effect	Full-stop	90% interval	Years	p-value	
(km/h)		(×100)	minutes	(minutes)	negative	single	stepdown
All cells	2 305	−0.30	0.18	−0.07 to 0.43	3 of 5	0.116	–
< 30	518	−0.06	0.04	−0.42 to 0.47	2 of 5	0.441	0.718
30–50	608	−0.44	0.26	−0.12 to 0.65	3 of 5	0.133	0.328
50–70	391	−1.07	0.64	0.26 to 1.05	5 of 5	0.003	0.015
≥ 70	788	−0.02	0.01	−0.32 to 0.36	3 of 5	0.486	0.718

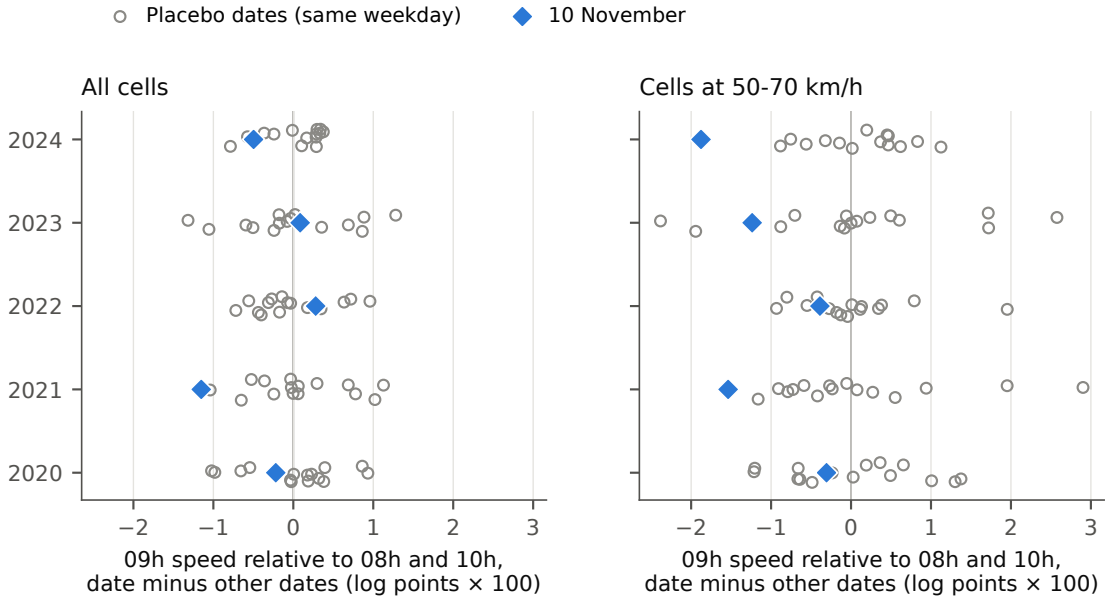


Figure 2: The statistic $T_y(d)$ for each date of each year's pool: speed in the 09:00 hour relative to the 08:00 and 10:00 hours, date minus the other dates of the year. Open circles are placebo dates of the same weekday (vertical position jittered); diamonds are 10 November.

Table 8).

Figure 3 shows the estimate in ten-km/h bands. It is largest in the 50–60 band (0.84 minutes, $p = 0.001$), smaller and not significant by itself in the 60–70 band (0.42 minutes, $p = 0.089$), and within the range of placebo dates everywhere else.

5.2 Minimum speed, maximum speed and vehicles

If vehicles brake to a halt and restart, the slowest observation in the hour should be slower, the fastest should not change, and the number of distinct vehicles should not change. Table 3 shows this pattern. The minimum speed in the 09:00 hour fell by 4.7% in the 50–70 class ($p = 0.015$) and by 3.4% over all cells ($p = 0.026$, negative in 4 of 5 years). The maximum speed and the number of vehicles did not move in either group. In 2021–2024, when the threshold was 1 km/h, the share of 50–70 cells whose minimum speed was 2 km/h or less rose by 1.9 percentage points

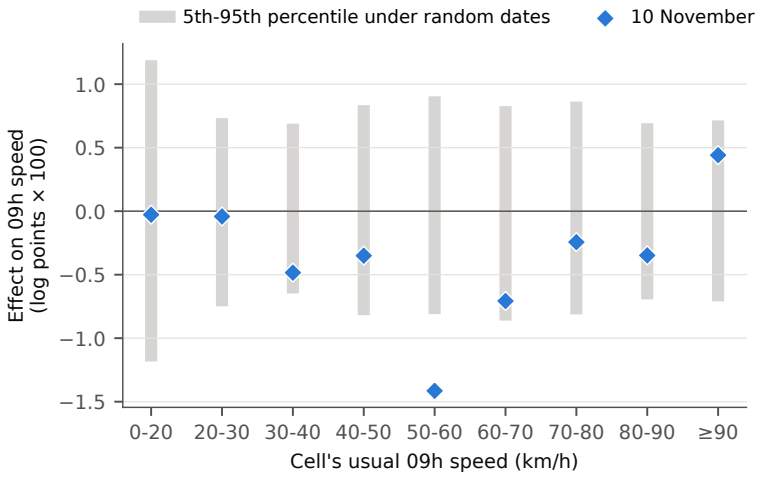


Figure 3: Effect on speed in the 09:00 hour by the cell's usual speed at 09:00. Diamonds are the estimates for 10 November; grey bars span the 5th to 95th percentile of the same statistic over assignments of one date per year. One log point ×100 is about 0.6 full-stop-equivalent minutes.

in the 09:00 hour ($p = 0.014$).

5.3 Other hours and other days

The same contrast can be formed for any hour: speed in hour h relative to hours $h-1$ and $h+1$. Figure 4 shows it for the 50–70 class. The 09:00 hour gives -0.93 ($p = 0.012$). The 08:00 and 10:00 values are raised mechanically, since each uses the 09:00 hour as a reference. The 10:00 value (1.02) is higher than this mechanical share alone would give, which indicates that the 10:00 hour was itself fast on 10 November. The 09:00 hour is not alone: of the 22 hours, 3 (06, 09 and 18) fall below the 5th percentile of their placebo distribution, against one expected by chance. Contrasts centred on 07:00 and 11:00, which do not involve the 09:00 hour, are not significant in the 50–70 class ($p = 0.621$ and 0.127); over all cells the 11:00 contrast is negative ($p = 0.030$). Moving the "event" to 9 or 11 November gives nothing (Table 3).

5.4 Which hour moves

The contrast compares the 09:00 hour with its two neighbours, so a fast neighbour and a slow 09:00 hour look alike. To separate them we measure each of the three hours against the mean of the hours 06, 07, 11 and 12, none of which adjoins the siren. In the 50–70 class the 08:00 hour is ordinary on 10 November (-0.12, $p = 0.424$). The 10:00 hour is fast: $+1.05$ log points ($\times 100$), positive in 4 of 5 years, with a one-sided p-value of 0.032 in that direction. The 09:00 hour is slow by -0.73, or 0.43 minutes, negative in 3 of 5 years ($p = 0.101$). So the main estimate has two parts: a 09:00 hour that is slower than distant hours by about 0.43 minutes, and a 10:00 hour that is faster than usual. The minimum speed shows the same split. Against the 08:00 hour alone it is lower at 09:00 by 1.8% ($p = 0.256$); against the 10:00 hour alone, by 7.5% ($p = 0.003$; Table 3). The fast 10:00 hour has two candidate explanations: it belongs to the event, as a release after the stop or as lighter traffic after the ceremonies, or it is a feature of the day unrelated to the siren.

Table 3: Other outcomes and placebo tests. Each row applies the main design to another outcome, hour or date. Effects are in log points ×100, except the share of cells (percentage points). "Years" counts years with a negative estimate (positive for the share). p-values are one-sided in the direction of a slowdown.

	All cells			Cells at 50–70 km/h		
Outcome	Effect (×100)	p	Years	Effect (×100)	p	Years
Minimum speed (log)	−3.49	0.026	4 of 5	−4.81	0.015	5 of 5
reference hour 08 only	−1.40	0.282	4 of 5	−1.77	0.256	3 of 5
reference hour 10 only	−5.53	0.004	5 of 5	−7.78	0.003	5 of 5
Maximum speed (log)	0.04	0.540	2 of 5	−0.04	0.483	3 of 5
Number of vehicles (log)	0.03	0.496	2 of 5	−0.39	0.468	2 of 5
Share of cells with minimum speed ≤ 2 km/h	0.66	0.217	3 of 4	1.92	0.014	3 of 4
Placebo hour 07 (vs 06 and 08)	−0.28	0.244	4 of 5	0.19	0.621	2 of 5
Placebo hour 11 (vs 10 and 12)	−0.38	0.030	4 of 5	−0.43	0.127	3 of 5
Placebo event on 9 November	0.03	0.477	2 of 5	−0.17	0.443	3 of 5
Placebo event on 11 November	−0.33	0.158	3 of 4	−0.20	0.345	3 of 4

5.5 Regression with fixed effects and weather

Table 4 gives the regression version. With precision-weighted cells, the 09:00 hour in the 50–70 class is slower than its neighbours by 0.57 minutes, negative in all five years ($p = 0.002$). Against hours that do not adjoin the siren it is slower by 0.41 minutes, again in all five years ($p = 0.062$), and against the 08:00 hour by 0.28 ($p = 0.174$). Over all cells the estimates stay small (0.05 minutes, $p = 0.376$).

The lower panel of the table compares estimators for the 50–70 class. Unweighted cells give 0.55 minutes ($p = 0.010$); dropping the weather controls gives 0.61. Fitting the fixed effects and weather by Huber M-estimation and then averaging the residuals plainly, as Rosenbaum (2002a) does, gives 0.56 ($p = 0.009$). Applying the Huber weights to every residual, including those on the event dates, gives 0.24 ($p = 0.040$). That version trims the event: among 09:00 cell-hours slower than predicted, Huber down-weights 15.8% on 10 November against 13.4% on placebo dates. Precision weighting also gives the best-behaved residuals: their kurtosis is 11, against 22 for unweighted cells and 38 for the Huber fit (Appendix Figure 7).

5.6 Where

Within the 50–70 class the estimate is 0.66 minutes on the European side and 0.64 on the Asian side, and the difference is far from significant ($p = 0.969$; Table 6). Splitting the cells by the district's vote for the Republican People's Party (CHP) candidate in the 2024 mayoral election gives 0.64, 0.88 and 0.51 minutes from the lowest to the highest band of vote share, with no significant difference between groups (smallest $p = 0.450$). Across the 19 districts with enough cells, the estimate is negative in 14 and its correlation with the CHP share is −0.05 (Appendix Figure 6). Cells with few and many probe vehicles give similar estimates.

On slower roads the vehicles in a cell are more likely to belong to the surrounding area, so the local vote is a closer proxy for the drivers there than on arterials and motorways that pass through. Table 5 reports, for each speed class, the slope of the cell-level response on the CHP share (standardized over cells), from the regression residuals, with placebo-date p-values, once

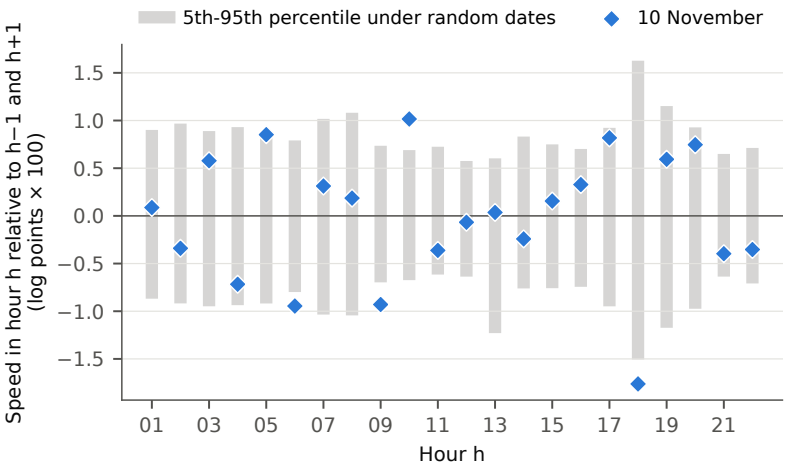


Figure 4: Speed in each hour relative to the hours before and after it, 10 November minus placebo dates, cells at 50–70 km/h. Diamonds are the estimates; grey bars span the 5th to 95th percentile over assignments of one date per year. Hours 08 and 10 are raised mechanically by the dip at 09.

with the district share and once with the neighbourhood share.

Neighbourhood shares, which match the size of a cell, show a clear gradient on slow streets. With the 2023 parliamentary vote, cells below 30 km/h in neighbourhoods with a higher CHP share slow more against hours that do not adjoin the siren, in all five years: by −1.08 log points (×100) per standard deviation ($p = 0.0007$, or 0.007 adjusted for the ten slopes tested). Against the neighbouring hours the slope is negative in 4 of 5 years ($p = 0.009$). The 2024 mayoral vote gives the same pattern, slightly stronger and negative in all five years ($p = 0.0004$ and 0.001). Over all cells the parliamentary-vote slope against distant hours is negative in all five years ($p = 0.032$). In the 50–70 class the slopes are negative but not significant, and motorway cells show no gradient. District shares give a weaker version of the slow-street result (1 of ten slopes significant, $p = 0.011$), because districts mix neighbourhoods of different leanings. The European and Asian sides do not differ in any class (smallest $p = 0.418$).

A standard deviation of the neighbourhood CHP share is 12.1 percentage points in 2023. In slow cells the average response is close to zero, so the gradient implies a slowdown at 09:00 in neighbourhoods with a high CHP share and a small speed-up, or none, in neighbourhoods with a low one.

5.7 Comparison with a city-wide index

A single city-wide series hides the disturbance that the cell-level design finds. Figure 5 shows İBB's traffic index at five-minute intervals on the morning of 10 November 2026, a Tuesday, against the other 35 working days of the preceding two months. The index averages over the whole network, in which the cells that flow at 50–70 km/h are 17% of all cells, and it shows no rise at 09:05. The statistic analogous to the hourly contrast, the index in the two five-minute readings after 09:00 minus the mean of the readings twenty minutes earlier and later, is 1.5 index points on 10 November, against a mean of −0.2 and a standard deviation of 1.3 on the other days; it ranks 5 of 36 ($p = 0.139$). We chose this window after seeing the series; windows of fifteen and thirty minutes give ranks of 5 and 7. Grouping cells by their usual speed is what isolates the arterial roads on which the 09:00 hour is slow, and a finer time step applied to an

Table 4: Fixed-effects regression version. Cell-by-date and cell-by-day-type-by-hour fixed effects, school-break hour effects, and hourly rain and temperature are partialled out over all days within eight weeks of 10 November; the event statistic is formed from the residuals of same-weekday pool dates. In the main version each cell is weighted by the inverse of its residual variance on days outside the pool. Minutes are full-stop-equivalent minutes; p-values are one-sided placebo-date p-values.

	09 vs 08 and 10		09 vs 06, 07, 11, 12		09 vs 08	
Usual 09h speed (km/h)	Minutes	p	Minutes	p	Minutes	p
Precision-weighted cells (main version)						
All cells	0.05	0.376	−0.03	0.553	−0.17	0.792
< 30	−0.04	0.561	0.29	0.215	−0.46	0.890
30–50	0.24	0.115	0.24	0.243	−0.37	0.832
50–70	0.57	0.002	0.41	0.062	0.28	0.174
≥ 70	−0.07	0.640	−0.26	0.869	−0.12	0.653
Cells at 50–70 km/h, other estimators						
Unweighted cells	0.55	0.010	0.26	0.219	0.28	0.238
Unweighted, no weather controls	0.61	0.005	0.31	0.179	0.32	0.206
Huber background fit, plain mean	0.56	0.009	0.27	0.211	0.29	0.228
Full Huber (weights on every residual)	0.24	0.040	0.10	0.305	−0.02	0.536

average over all roads does not replace it.

6 Robustness

Table 7 repeats the analysis under other choices, for all cells and for the 50–70 class. Of the 29 specifications for the 50–70 class, 26 give a p-value below 0.05. The three that do not are the three that avoid the 10:00 hour as a reference.

Placebo-date pool. With the three control dates of the pilot (3, 17 and 24 November) the 50–70 estimate is 0.56 minutes ($p = 0.009$), and with the two adjacent weeks alone 0.56 ($p = 0.012$). Pools of four, eight and twelve weeks on each side give 0.90, 0.64 and 0.73 minutes, every same-weekday date of the calendar year gives 0.52 ($p = 0.043$), and keeping the school-break dates gives 0.71. Across these pools the estimate lies between 0.52 and 0.90 minutes and the p-value never exceeds 0.043. In contrast with the election-day and 20 April studies (Zhang and Aronow, 2016; Harper and Palayew, 2019), widening the pool leaves the result for this class in place. It removes the result for the 30–50 class, whose pilot p-value of 0.034 becomes 0.133 (0.328 after adjustment).

Trend and scale. Removing a linear or quadratic trend in M_{yd} across the weeks of each year, fitted without the date under test, gives 0.64 and 0.77 minutes. A raw difference in means can mislead when the event dates are more variable than the others (Chung and Romano, 2013; Wu and Ding, 2020). Dividing each year's statistics by the standard deviation of its placebo dates gives $p = 0.002$, and replacing them by ranks gives $p = 0.005$. Removing the part of M_{yd} explained by rain and temperature, in the manner of Rosenbaum (2002a), gives 0.59 minutes.

Reference hours. This is where the estimate is fragile. Against the 08:00 hour alone it is 0.32 minutes ($p = 0.208$); against the 10:00 hour alone, 0.93 ($p = 0.008$). Against the four

Table 5: Political gradient. Slope of the cell-level event response on the CHP vote share (standardized over cells): district share in the 2024 mayoral election, and neighbourhood shares in the 2023 parliamentary and 2024 mayoral elections, by usual 09:00 speed, in log points ×100 per standard deviation. A negative slope means a larger slowdown where the CHP share is higher. Superscripts count the years (of five) with a negative slope. p-values are two-sided placebo-date p-values. The neighbourhood share of a cell is area-weighted over the neighbourhoods it overlaps.

	District, 2024 local		Neighbourhood, 2023 general		Neighbourhood, 2024 local	
Usual 09h speed (km/h)	Slope	p	Slope	p	Slope	p
Panel A: 09:00 against 08:00 and 10:00						
All cells	-0.03^{2}	0.791	-0.05^{3}	0.747	-0.04^{3}	0.809
< 30	-0.19^{4}	0.292	-0.60^{4}	0.009	-0.72^{5}	0.001
30–50	-0.01^{2}	0.956	-0.01^{2}	0.980	-0.04^{2}	0.895
50–70	-0.27^{3}	0.370	-0.53^{4}	0.227	-0.66^{4}	0.169
≥ 70	0.15^{2}	0.604	0.27^{2}	0.290	0.36^{1}	0.186
Panel B: 09:00 against 06:00, 07:00, 11:00 and 12:00						
All cells	-0.30^{4}	0.093	-0.37^{5}	0.032	-0.31^{4}	0.085
< 30	-0.52^{5}	0.011	-1.08^{5}	0.0007	-1.19^{5}	0.0004
30–50	-0.22^{4}	0.481	-0.12^{4}	0.723	-0.21^{4}	0.540
50–70	-0.23^{3}	0.482	-0.70^{5}	0.106	-0.69^{4}	0.144
≥ 70	-0.17^{4}	0.514	-0.03^{2}	0.888	0.13^{2}	0.624

hours 07, 08, 10 and 11 it is 0.56 ($p = 0.015$), and against the hours 06, 07, 11 and 12, none of which adjoins the event, 0.43 ($p = 0.101$). Subtracting the mean contrast at the placebo hours 07 and 11 gives 0.67 ($p = 0.014$). The contrast against the earlier hour is the cleanest in principle, because the later hour comes after the event (Rosenbaum, 2002b); it is also the weakest. A fast hour after a synchronized pause has been seen elsewhere (Olexsak and Meier, 2014; Ngeni et al., 2022), but here it comes a full hour later, and we do not attribute it to the stop.

Cells and classification. Restricting to cells observed on every pool date, or to cells with at least ten probe vehicles, gives 0.72 and 0.74 minutes. Weighting cells by their number of probe vehicles gives 0.62 minutes ($p = 0.046$). An earlier version of the analysis classified cells by their speed on the placebo dates only; it gave a somewhat larger estimate and was replaced, because it treats the event date differently from the others. Classifying cells by their speed on days outside the pool, so that classification and estimation share no data, gives a smaller estimate of 0.51 minutes ($p = 0.014$), negative in all five years. Some of the difference is the dilution that comes with a noisier classification; some may be selection on the same noisy speeds that enter the contrast, and we regard this estimate as the more cautious one.

Years. Leaving out one year at a time gives between 0.52 and 0.75 minutes, with p-values of at most 0.030. Without 2024, when the event fell on a Sunday, the estimate is 0.52 minutes. Without 2020, the year of the higher speed threshold and of pandemic restrictions, it rises to 0.75.

Sensitivity and power. Following Rosenbaum (2002b), we ask how far the uniform benchmark must fail before the conclusion changes. If within each year the odds of being the event date could differ between dates by a factor Γ, the worst-case p-value for the 50–70 class stays

Table 6: Estimates for groups of cells within the 50–70 km/h class.

Group	Cells	Effect (×100)	Minutes	Years negative	p
Side of the Bosphorus					
European side	230	−1.11	0.66	4 of 5	0.009
Asian side	161	−1.08	0.64	4 of 5	0.050
CHP vote share 2024 (district)					
CHP < 45%	75	−1.07	0.64	4 of 5	0.090
CHP 45–52%	112	−1.48	0.88	4 of 5	0.025
CHP > 52%	204	−0.85	0.51	4 of 5	0.040
Probe vehicles at 09h					
< 30 vehicles	131	−0.72	0.43	4 of 5	0.139
30–100 vehicles	129	−1.29	0.77	5 of 5	0.008
> 100 vehicles	131	−1.26	0.75	5 of 5	0.032

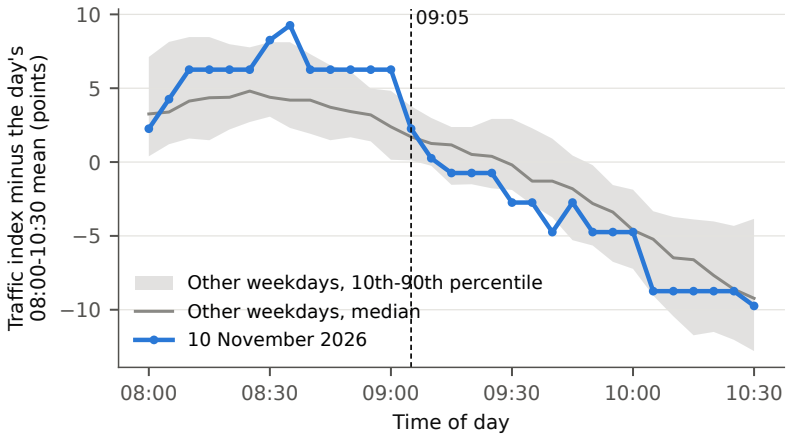


Figure 5: İBB city-wide traffic index (higher is more congested) at five-minute intervals on 10 November 2026 and on the other working days from 14 September to 9 November 2026, each day measured from its own mean between 08:00 and 10:30.

below 0.05 up to $\Gamma = 3$. To gauge what the design can detect, we lowered the statistic of one randomly chosen placebo date per year by the effect of a full stop of given length and reran the test. A stop equivalent to half a minute is detected at the 5% level in 96% of simulations over all cells and 68% in the 50–70 class; one of a quarter of a minute in 49% and 29%. The small estimate over all cells is therefore informative: a measured slowdown of half a minute across all cells would very probably have been found.

7 Discussion

What the data show. On roads that normally flow at 50–70 km/h, the hourly profile of 10 November is unusual around the siren in every one of five years: the 09:00 hour lies below the line between its neighbours by an amount that few random choices of dates would produce. The 09:00 hour is slower than more distant hours by about 0.43 minutes, the minimum speed falls and the maximum does not, as braking and restarting would imply. The rest of the gap comes from a fast 10:00 hour. We read this as evidence of a disturbance at the siren on these roads, with a size between 0.43 and 0.64 full-stop-equivalent minutes. The pattern does

Table 7: Robustness. Each row changes one feature of the main specification. Minutes are full-stop-equivalent minutes; p-values are one-sided placebo-date p-values; "Years < 0" counts years with a negative estimate.

	All cells			Cells at 50–70 km/h		
Specification	Minutes	p	Years < 0	Minutes	p	Years < 0
Placebo-date pool						
Main specification (±8 weeks)	0.18	0.116	3 of 5	0.64	0.003	5 of 5
Pilot dates only (3, 17, 24 Nov)	0.21	0.067	3 of 5	0.56	0.009	5 of 5
Adjacent weeks only (3 and 17 Nov)	0.17	0.128	3 of 5	0.56	0.012	5 of 5
±4 weeks	0.18	0.065	3 of 5	0.90	0.0002	5 of 5
±12 weeks	0.15	0.180	3 of 5	0.73	0.0004	5 of 5
All same-weekday dates of the year	0.08	0.327	3 of 5	0.52	0.043	5 of 5
School-break dates kept	0.19	0.100	3 of 5	0.71	0.0006	5 of 5
Trend and scaling						
Linear trend removed within year	0.18	0.111	3 of 5	0.64	0.002	5 of 5
Quadratic trend removed within year	0.20	0.098	3 of 5	0.77	0.0008	5 of 5
Statistic scaled by placebo s.d.	–	0.098	3 of 5	–	0.002	5 of 5
Rank statistic	–	0.164	–	–	0.005	–
Adjusted for rain and temperature	0.16	0.135	3 of 5	0.59	0.004	5 of 5
Reference hours						
Reference hour 08 only	−0.13	0.691	3 of 5	0.32	0.208	3 of 5
Reference hour 10 only	0.49	0.029	3 of 5	0.93	0.008	4 of 5
Reference hours 07, 08, 10, 11	0.13	0.241	3 of 5	0.56	0.015	5 of 5
Reference hours 07 and 11	0.07	0.390	2 of 5	0.52	0.072	4 of 5
Reference hours 06, 07, 11, 12	0.19	0.245	3 of 5	0.43	0.101	3 of 5
Net of curvature at placebo hours 07 and 11	0.02	0.469	1 of 5	0.67	0.014	5 of 5
Cells						
Balanced cells (observed on every date)	0.23	0.064	4 of 5	0.72	0.002	5 of 5
Cells with ≥ 10 vehicles at 09h	0.18	0.124	3 of 5	0.74	0.0010	5 of 5
Cells weighted by probe vehicles	0.17	0.239	3 of 5	0.62	0.046	4 of 5
Speed class from other days	–	–	–	0.51	0.014	4 of 5
Years						
Without 2020	0.19	0.127	2 of 4	0.75	0.002	4 of 4
Without 2021	0.05	0.369	2 of 4	0.57	0.015	4 of 4
Without 2022	0.27	0.065	3 of 4	0.74	0.004	4 of 4
Without 2023	0.24	0.066	3 of 4	0.61	0.003	4 of 4
Without 2024	0.15	0.202	2 of 4	0.52	0.030	4 of 4
Weekday events only (2020–2023)	0.15	0.202	2 of 4	0.52	0.030	4 of 4
1 km/h floor years only (2021–2024)	0.19	0.127	2 of 4	0.75	0.002	4 of 4

not depend on the side of the city. At the level of districts it does not depend on politics either, which fits Goldman's (2018) account of a commemoration shared across camps. At the level of neighbourhoods it does: on slow, local streets, the slowdown rises with the CHP vote, the secular profile that Özyürek (2004) associates with voluntary Atatürk commemoration. Districts mix neighbourhoods of different leanings, and arterial traffic mixes drivers from across the city; the neighbourhood gradient on local streets is the comparison in which drivers and voters coincide most closely.

Why only there. The estimate is absent below 50 km/h and above 70 km/h. Studies of demand shocks find larger effects on more congested roads (Adler et al., 2021; Hanna et al., 2017); we find the opposite at the slow end. The measurement explains part of this. Where traffic already stops and starts, standing is routine, the average is already a running speed, and a further stop adds little that the data retain. At the fast end, cells with a usual speed above 70 km/h are motorways, where stopping in a live lane is dangerous and may be rarer. Lower participation and weaker measurement both predict this pattern.

What the minutes mean. If the published average were distance over time for all vehicles, half a minute would be the standstill per vehicle: half of the vehicles stopping for one minute, or fewer for longer. The threshold in the data rules this reading out. If instead only moving vehicles are averaged, full compliance with ordinary braking would produce between about 0.15 and 0.35 minutes. Our estimates, from 0.32 to 0.93 minutes depending on the reference hour, lie at the top of that range or above it, so they contain more than clean stops: crawling vehicles and slow restarts of platoons enter, and so, in the main contrast, does the fast 10:00 hour. The smaller estimate for 2020 (0.18 minutes), when observations below 6 km/h were discarded, is in line with this; 2020 was also a year of pandemic restrictions. In comparable units the effect is small: about 1% of travel time in the hour, a tenth or less of what a one-day transit strike adds in the morning peak (Bauernschuster et al., 2017; Anderson, 2014).

Scope. The estimate rests on five events and on one of four speed classes, chosen in a pilot on the same five event dates; the wider pool of placebo dates is the new evidence. It varies with the reference hour between 0.32 and 0.93 minutes, and the versions that avoid the 10:00 hour have p-values between 0.101 and 0.208. Ceremonies at 09:05 in schools and offices shift trips on the day. The estimates are full-stop-equivalent minutes of measured speed, a lower bound on the delay per vehicle.

Extension. The same placebo-date design applies directly to speeds recorded every minute on arterial segments from 08:45 to 09:30 on 10 November and on comparison days. İBB's own sensors record at this resolution (Takak et al., 2021). At that resolution the stop appears as a dip of measured depth and length, and the release after it is observed directly.

8 Conclusion

In hourly open data, Istanbul's remembrance stop leaves a trace only where traffic normally flows freely on arterial roads. There the 09:00 hour of 10 November lies below its neighbours in every year, by the equivalent of 0.64 minutes of full stop per vehicle; measured against more distant hours the figure is 0.43 minutes and is no longer statistically distinguishable from an

ordinary date. Both are lower bounds, because standing vehicles are missing from the data. Averaged over the whole city the hour is not measurably slower than on any other day, and the design would probably have detected half a minute. The design that finds this, a within-day contrast judged against a wide pool of placebo dates, guards against the failures documented for adjacent-week comparisons, and it locates the disturbance on the roads where it occurs, which an average over the whole city hides. Minute-level speeds on the same roads on the next 10 November extend the design to the depth and length of the stop.

Data and code

All data are public. Code to download the data and reproduce every number, table and figure is at `https://github.com/msamiboz/saygi-durusu-traffic`. Numbers in the text are generated by the code.

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Table 8: Yearly estimates $T_y(e_y)$ by speed class, log points ×100.

Usual 09h speed (km/h)	2020	2021	2022	2023	2024
All cells	−0.22	−1.15	0.28	0.09	−0.50
< 30	0.80	−2.42	1.16	1.29	−1.15
30–50	−0.23	−1.50	0.17	0.08	−0.72
50–70	−0.31	−1.54	−0.39	−1.24	−1.88
≥ 70	−0.84	0.32	−0.01	−0.17	0.60

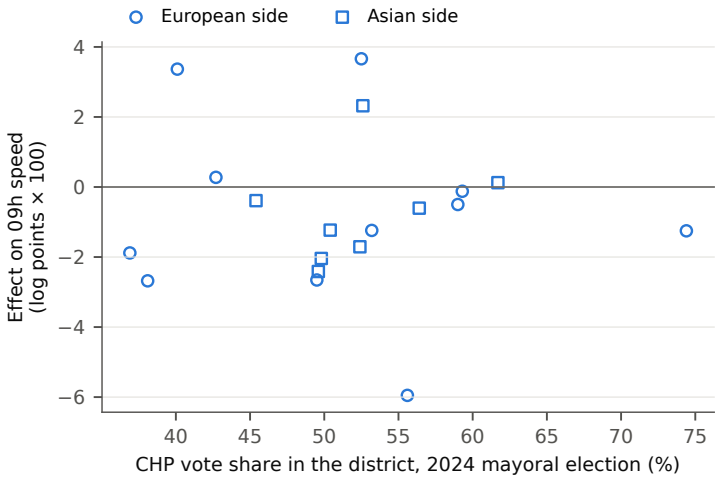


Figure 6: District estimates for cells at 50–70 km/h against the district's vote share for the CHP candidate in the 2024 metropolitan mayoral election. Circles are districts on the European side, squares on the Asian side. Districts with fewer than five such cells in any year are omitted.

A Yearly estimates

B What an hourly mean retains of a brief stop

Consider one cell and one hour with baseline speed v_0. A share π of vehicles brakes linearly for t_d minutes, stands for s minutes and accelerates linearly for t_a minutes. Each loses the distance $v_0 k^*$, where $k^* = s + (t_d + t_a)/2$ is the full-stop-equivalent duration.

(A) Speeds sampled at fixed time intervals, all samples kept. Samples are proportional to vehicle-time, so the mean is total distance over total time, $v_0(1-\pi k^*/60)$, and $60(1-e^D) = \pi k^*$.

(B) Fixed intervals, stationary samples discarded. The same distance is divided by moving time, $60 - \pi s$. The mean is $v_0(60 - \pi k^*)/(60 - \pi s)$ and $60(1 - e^D) \approx \pi(t_d + t_a)/2$.

(C) Speeds recorded on crossing a point. The mean is flow-weighted. By the identity $v_t = v_s + \sigma_s^2/v_s$ of Wardrop (1952), a population that spends a fraction $s/60$ of the hour at zero speed and the rest at v_0 has space-mean speed $v_0(1 - s/60)$ and time-mean speed exactly v_0. Only the ramps register, with a loss of about $\pi(t_d + t_a)/6$.

(D) One speed per traversal of the cell, averaged over vehicles. If a free traversal takes τ minutes, a share $\tau/60$ of the hour's traversals meets the stop, each at speed $v_0\tau/(\tau + k^*)$, so $60(1 - e^D) \approx \pi\tau k^*/(\tau + k^*)$, which is below both τ and k^*.

Scheme A predicts the same minutes in every speed class; B predicts minutes proportional to speed, since $t_d + t_a$ is; D predicts more minutes in slower cells. The data show none of these

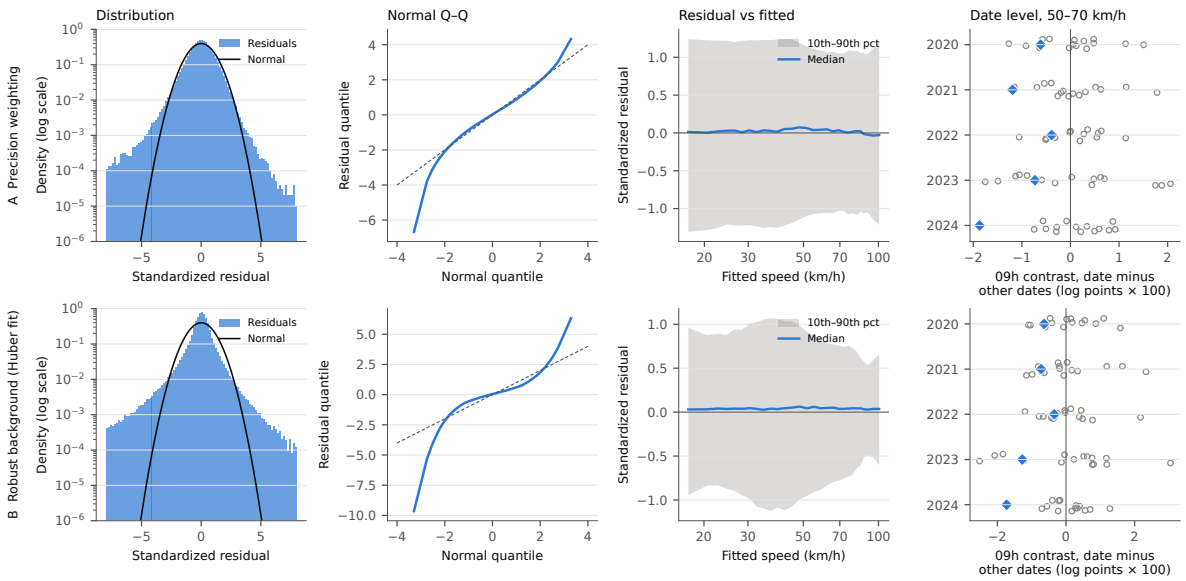


Figure 7: Residual diagnostics of the fixed-effects regression. Top row: residuals standardized by each cell's own residual s.d. on days outside the pool (precision weighting, the main version). Bottom row: residuals of the Huber fit, scaled by the Huber scale. Columns: distribution against a normal density (log scale); normal Q–Q plot; residuals against fitted speed (median and 10th–90th percentiles); the date-level 09:00 contrast for cells at 50–70 km/h, 10 November (diamonds) against same-weekday placebo dates (circles).

gradients (Figure 3), a minimum speed that is never zero, an unchanged number of vehicles, and a lower minimum speed in the event hour. Schemes B and C, with some crawling vehicles, are the readings most in line with these facts.